Siddhartha Prasad

siddhartha.a.prasad@gmail.com | siddharthaprasad.com

I am a programming languages researcher interested in the human aspects of programming and formal reasoning. I study not only how PL and FM techniques can solve problems programmers face, but also how programmers' experiences can inform the design of our languages, logics, and software tools. My research is informed by my time as an engineer: I have written code that doesn't do what I want it to, and I want to spare everyone else the indignity.

Education

Ph.D. in Computer Science, **Brown University**, Providence, RI *Jan 2022 – present*

Advisor: Shriram Krishnamurthi

B.S. in Computer Science and Mathematics, **Tufts University**, Medford, MA *May 2016*

Advisor: Kathleen Fisher · *Summa cum Laude; High Thesis Honors*

Employment

Apple · Research Intern · Human-Centered Machine Learning · Seattle, WA *May – Sept 2023*

Designed a formal-methods-inspired technique that prompts an LLM to evaluate candidate outputs against task-specific quality criteria and iteratively refine them without fine-tuning. The technique drives ContextQ, a system that generates questions to support parent-child dialogue during shared reading.

Microsoft Redmond, WA

Software Engineer II · Applied AI *April 2018 – Sept 2021*

- Built the container platform for Azure Cognitive Services, enabling customers with sensitive data to run AI workloads on-premises, at the edge, or in fully disconnected environments.
- Designed a deployment framework that allowed the same Docker containers to run across Azure-hosted and customer-managed environments, abstracting differences in CPU architecture, connectivity, API-key validation, logging, and storage.
- Operated production infrastructure for the cloud service, including on-call responsibility.

Software Engineer · Developer Ecosystem Platform *Aug 2016 – March 2018*

- Designed and shipped XAML input APIs across Windows 10, Windows Phone, and Xbox, giving developers a consistent model for keyboard, gamepad, and ink interaction.
- Built accessibility and navigation features, including XAML access keys, so applications could expose commands and behave consistently across keyboard and gamepad input.

Software Engineering Intern · Developer Ecosystem Platform *May – Aug 2015*

INRIA · Research Intern · Parsifal · Saclay, France *June – July 2014*

Research on correctness certificates for first-order term-rewriting systems.

Selected Publications

Diagramming Program Values by Spatial Refinement, **PLDI 2026**
Siddhartha Prasad, Michael Tu, Karan Kashyap, Tim Nelson, Shriram Krishnamurthi

Meaningful Human-in-the-Loop Checking of GenAI Synthesis for Restricted Languages, **ECOOP 2026**
Siddhartha Prasad, Skyler Austen, Kathi Fisler, Shriram Krishnamurthi

Lightweight Diagramming for Lightweight Formal Methods: A Grounded Language Design
Distinguished Paper and Distinguished Artifact at **ECOOP 2025**
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi

A Misconception-Driven Adaptive Tutor for Linear Temporal Logic
Distinguished Paper at **CAV 2025**
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi

Misconceptions in Finite-Trace and Infinite-Trace Linear Temporal Logic, **FM 2024**
Ben Greenman, *Siddhartha Prasad*, Antonio Di Stasio, Shufang Zhu, Giuseppe De Giacomo, Shriram Krishnamurthi, Marco Montali, Tim Nelson, Milda Zizyte

Forge: A Tool and Language for Teaching Formal Methods, **OOPSLA 2024**
Tim Nelson, Ben Greenman, *Siddhartha Prasad*, Tristan Dyer, Ethan Bove, Qianfan Chen, Charles Cutting, Thomas Del Vecchio, Sidney LeVine, Julianne Rudner, Ben Ryjikov, Alexander Varga, Andrew Wagner, Luke West, Shriram Krishnamurthi